

27 Supersymmetric Gauge Theories

27.5 Supersymmetry Breaking in the Tree Approximation Resumed

We saw in the previous section that if the Fayet–Iliopoulos constants ξ_A all vanish and if there exists a solution of the equations $\partial f(\phi)/\partial\phi_n = 0$, then there is also some solution of these equations where the D -components of the gauge superfields all vanish, so that supersymmetry is unbroken. It follows that there are only two (non-exclusive) ways that supersymmetry can be spontaneously broken in the tree approximation in renormalizable theories of gauge and chiral superfields: the superpotential $f(\phi)$ may be arranged so that there are no solutions of all the equations $\partial f(\phi)/\partial\phi_n = 0$, or for gauge groups with $U(1)$ factors there may be Fayet–Iliopoulos terms in the action.

We have already seen in Section 26.5 how it can happen that there might not be any value of ϕ for which $\partial f(\phi)/\partial\phi_n = 0$. No change in that discussion is needed when the chiral superfields interact with gauge fields, so let's turn to the other possibility: spontaneous supersymmetry breaking produced by Fayet–Iliopoulos terms. Since this can only arise for gauge groups with $U(1)$ factors, the simplest case is a theory with a single $U(1)$ gauge group. As discussed in Section 22.4, to avoid $U(1)$ - $U(1)$ - $U(1)$ and $U(1)$ -graviton-graviton anomalies it is necessary that the sum of the $U(1)$ quantum numbers of all left-chiral superfields and the sum of their cubes should vanish. We will consider the simplest possibility: two left-chiral superfields $\Phi_{\pm}$, with $U(1)$ quantum numbers $\pm e$. (This is a supersymmetric version of quantum electrodynamics, with the spinor components ψ_{-L} and ψ_{+L} of the two superfields providing the left-handed parts of the electron field and of its charge conjugate.) The most general $U(1)$ -invariant superpotential in a renormalizable theory is just $f(\Phi) = m\Phi_+\Phi_-$. The scalar potential (27.4.9) for the scalar components $\phi_{\pm}$ of these superfields is then

$$V(\phi_+, \phi_-) = m^2|\phi_+|^2 + m^2|\phi_-|^2 + (\xi + e^2|\phi_+|^2 - e^2|\phi_-|^2)^2. \quad (27.5.1)$$

Unless the Fayet–Iliopoulos constant ξ vanishes, it is evidently not possible to find a supersymmetric vacuum with $V = 0$. For $\xi > m^2/2e^2$ or $\xi < -m^2/2e^2$ the potential (27.5.1) has a minimum with either $\phi_+ = 0$ and $|\phi_-|^2 = (2e^2\xi - m^2)/2e^4$ or $\phi_- = 0$ and $|\phi_+|^2 = (-2e^2\xi - m^2)/2e^4$, so the $U(1)$ gauge symmetry is broken along with supersymmetry. For $|\xi| < m^2/2e^2$ the minimum of the potential is at $\phi_+ = \phi_- = 0$, so here the gauge symmetry is unbroken. There is in general no necessary connection between the possible breakdown of supersymmetry and of gauge symmetries.

Whether supersymmetry is spontaneously broken by the Fayet–Iliopoulos mechanism discussed here or by the O’Raifeartaigh mechanism of Section 26.5 or by some combination of the two, supersymmetry leaves a remnant in the pattern of tree-approximation masses. Inspection of the Lagrangian (27.4.8) for a general renormalizable supersymmetric theory of gauge and chiral superfields shows that the spontaneous breakdown of supersymmetry in this theory produces the following corrections to the masses calculated in Section 27.4.

Spin 0 Masses

If the $\mathcal{F}$ -terms $\mathcal{F}_n = -(\partial f(\phi)/\partial\phi_n)^*$ do not vanish at the minimum ϕ_0 of the potential, then the terms in the potential of second order in $\varphi_n \equiv \phi_n - \phi_{n0}$ have additional terms

beyond those listed in Eq. (27.4.15):

$$\begin{aligned}
V_{\text{quad}}(\varphi) = & \sum_{nm} (\mathcal{M}^* \mathcal{M})_{nm} \varphi_n^* \varphi_m + \sum_{Anm} (t_A \phi_0)_n (t_A \phi_0)_m^* \varphi_n^* \varphi_m \\
& + \frac{1}{2} \sum_{Anm} (t_A \phi_0)_n^* (t_A \phi_0)_m^* \varphi_n \varphi_m + \frac{1}{2} \sum_{Anm} (t_A \phi_0)_n (t_A \phi_0)_m \varphi_n^* \varphi_m^* \\
& + \frac{1}{2} \sum_{nm} \mathcal{N}_{nm} \varphi_n \varphi_m + \frac{1}{2} \sum_{nm} \mathcal{N}_{nm}^* \varphi_n^* \varphi_m^* + \sum_{Anm} D_{A0} (t_A)_{nm} \varphi_n^* \varphi_m,
\end{aligned} \tag{27.5.2}$$

where $\mathcal{M}$ is again the complex symmetric matrix (26.4.11):

$$\mathcal{M}_{nm} \equiv \left(\frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} \right)_{\phi=\phi_0},$$

$\mathcal{N}_{nm}$ is a new ingredient

$$\mathcal{N}_{nm} \equiv - \sum_{\ell} \mathcal{F}_{\ell 0} \left(\frac{\partial^3 f(\phi)}{\partial \phi_n \partial \phi_m \partial \phi_{\ell}} \right)_{\phi=\phi_0}, \tag{27.5.3}$$

and $\mathcal{F}_0$ and D_{A0} are again the $\mathcal{F}$ -terms and D -terms of the chiral scalar and gauge superfields at the minimum of the potential:

$$\mathcal{F}_{n0} = - \left[\frac{\partial f(\phi)}{\partial \phi_n} \right]_{\phi=\phi_0}^*, \quad D_{A0} = \xi_A + \sum_{nm} \phi_{n0}^* (t_A)_{nm} \phi_{m0}.$$

If we write the quadratic part (27.5.2) of the potential in the form (27.4.16):

$$V_{\text{quad}} = \frac{1}{2} \begin{bmatrix} \varphi \\ \varphi^* \end{bmatrix}^{\dagger} M_0^2 \begin{bmatrix} \varphi \\ \varphi^* \end{bmatrix},$$

then, instead of Eq. (27.4.17), we now have the scalar mass matrix

$$M_0^2 = \begin{bmatrix} \mathcal{M}^* \mathcal{M} + \mathcal{A} + \sum_A D_{A0} t_A & \mathcal{B} + \mathcal{N}^* \\ \mathcal{B}^* + \mathcal{N} & \mathcal{M} \mathcal{M}^* + \mathcal{A}^* + \sum_A D_{A0} t_A^{\text{T}} \end{bmatrix}, \tag{27.5.4}$$

where

$$\mathcal{A} \equiv \sum_A (t_A \phi_0) (t_A \phi_0)^{\dagger}, \quad \mathcal{B} \equiv \sum_A (t_A \phi_0) (t_A \phi_0)^{\text{T}}.$$

Spin 1/2 Masses

The fermion mass matrix M here is given again by Eq. (27.4.30):

$$M_{nm} = \mathcal{M}_{nm}, \quad M_{nA} = M_{An} = i\sqrt{2} (t_A \phi_0)_n^*, \quad M_{AB} = 0.$$

However, now instead of Eq. (27.4.19), the gauge invariance condition (27.4.18) yields

$$\sum_m \mathcal{M}_{nm} (t_A \phi_0)_m = \sum_m \mathcal{F}_{m0}^* (t_A)_{mn}. \tag{27.5.5}$$

Thus the Hermitian positive matrix whose eigenvalues are the squared masses of the fermions is given by

$$\begin{aligned}(M^\dagger M)_{nm} &= (\mathcal{M}^\dagger \mathcal{M})_{nm} + 2 \sum_A (t_A \phi_0)_n (t_A \phi_0)_m^*, \\ (M^\dagger M)_{AB} &= 2(\phi_0^\dagger t_B t_A \phi_0), \\ (M^\dagger M)_{An} &= (M^\dagger M)_{nA}^* = i\sqrt{2} \sum_m \mathcal{F}_{m0}(t_A)_{mn}.\end{aligned}\tag{27.5.6}$$

Spin 1 Masses

The squared masses of the vector bosons are again given by the eigenvalues of the matrix (27.4.36):

$$(\mu^2)_{AB} = \left(\phi_0^\dagger \{t_B, t_A\} \phi_0 \right). \tag{27.5.7}$$

With the exception of the D -terms in Eq. (27.5.4), all of the changes in the mass-squared matrices are in their off-diagonal components. Therefore Eqs. (27.5.4), (27.5.6), and (27.5.7) yield particularly simple results for the traces of these matrices: for spin 0

$$\text{Tr } M_0^2 = 2 \text{Tr}(\mathcal{M}^* \mathcal{M}) + \text{Tr } \mu^2 + 2 \sum_A D_{A0} \text{Tr } t_A, \tag{27.5.8}$$

and for spin 1/2

$$\text{Tr}(M^\dagger M) = \text{Tr}(\mathcal{M}^* \mathcal{M}) + 2 \text{Tr } \mu^2. \tag{27.5.9}$$

Since the trace is the sum of the eigenvalues, we obtain from this a *mass sum rule*:

$$\sum_{\text{spin } 0} \text{mass}^2 - 2 \sum_{\text{spin } 1/2} \text{mass}^2 + 3 \sum_{\text{spin } 1} \text{mass}^2 = -2 \sum_A D_{A0} \text{Tr } t_A. \tag{27.5.10}$$

The trace of t_A automatically vanishes unless t_A is a $U(1)$ generator, and, as mentioned in Section 22.4, the trace must also vanish (when taken over all left-handed fermions) for $U(1)$ gauge generators to avoid gravitational contributions to an anomaly that would violate conservation of the $U(1)$ current. Thus (27.5.10) leads to the simpler result [4]

$$\sum_{\text{spin } 0} \text{mass}^2 - 2 \sum_{\text{spin } 1/2} \text{mass}^2 + 3 \sum_{\text{spin } 1} \text{mass}^2 = 0. \tag{27.5.11}$$

Of course, the unbroken conservation of charge, color, and baryon and lepton numbers prevents the mass matrices from having elements linking particles with different values of these quantum numbers, so all these results hold separately for each set of conserved quantum numbers.

The sum rule (27.5.11) is often quoted as providing evidence against models in which supersymmetry is spontaneously broken in the tree approximation within the minimum supersymmetric extension of the standard model. We will discuss this along with other arguments in Section 28.3.

As already observed in Section 26.5 (and discussed in greater generality in Sections 29.1 and 29.2), the spontaneous breakdown of supersymmetry necessarily entails the existence of a massless fermion, the goldstino. For renormalizable gauge theories in the tree

approximation, the goldstino field g appears as a term in the spinor components ψ_n and λ_A of the chiral and gauge superfields, with coefficients given by

$$\psi_{nL} = i\sqrt{2}\mathcal{F}_{n0}g_L + \cdots, \quad \lambda_{AL} = D_{A0}g_L + \cdots, \quad (27.5.12)$$

where the dots denote terms involving spinor fields of definite non-zero mass. To check this we have to confirm that $(i\sqrt{2}\mathcal{F}_{n0}, D_{A0})$ is an eigenvector of the fermion mass-squared matrix $M^\dagger M$ with eigenvalue zero. For this purpose we will need to use the condition that the potential (27.4.9) is stationary at $\phi = \phi_0$:

$$0 = \left. \frac{\partial V}{\partial \phi_n} \right|_{\phi=\phi_0} = - \sum_m \mathcal{M}_{nm} \mathcal{F}_{m0} + \sum_A D_{A0} (\phi_0^\dagger t_A)_n. \quad (27.5.13)$$

We also need the gauge invariance condition (27.4.12), which at $\phi = \phi_0$ reads

$$\sum_n \mathcal{F}_{n0} (t_A \phi_0)_n = 0. \quad (27.5.14)$$

Combining Eqs. (27.5.13) and (27.5.14) with Eqs. (27.5.5) and (27.5.6) then yields

$$\begin{aligned} i\sqrt{2} \sum_m (M^\dagger M)_{nm} \mathcal{F}_{m0} &= i\sqrt{2} \sum_A D_A (t_A \mathcal{F}_0^*)_n \\ &= - \sum_A (M^\dagger M)_{nA} D_{A0} \end{aligned} \quad (27.5.15)$$

and

$$\begin{aligned} i\sqrt{2} \sum_m (M^\dagger M)_{Am} \mathcal{F}_{m0} &= -2 \sum_{nm} \mathcal{F}_{n0} (t_A)_{nm} \mathcal{F}_{m0} \\ &= - \sum_B (M^\dagger M)_{AB} D_{B0}. \end{aligned} \quad (27.5.16)$$

That is,

$$M^\dagger M \begin{pmatrix} i\sqrt{2}\mathcal{F}_0 \\ D_0 \end{pmatrix} = 0, \quad (27.5.17)$$

as was to be shown.